

ICS Cybersecurity Evaluation (401) Training

Interdependencies Exercise



Create a System Map and Identify Dependencies.

Time: 15 minutes

Resources: Paper and pencil.

Instructions

Draw out a system map using the component items listed below (Feel free to add additional components to have more than one or to make up for something not listed here).

- Nuclear Power Plant
- Wind Farm
- Solar Panel Farm
- Hydropower Plant
- Transmission Substation
- Transmission Lines
- Distribution Substation
- Distribution Lines
- Residential Consumer
- Commercial Consumer
- Industrial Consumer

Draw out the path of power flow from sources to their recipients. Try to distinguish this power flow using a different color pen or line style (such as a dashed line).

Next, map out all the dependencies and interdependencies in the system using a different color or line style.

Consider: Even if you do not work in the power sector, think of it in broad terms. In what ways would an infrastructure rely on something else? Feel free to discuss this with your co-workers at your area of employment.

**This does
NOT need to
be submitted
after
completion.**

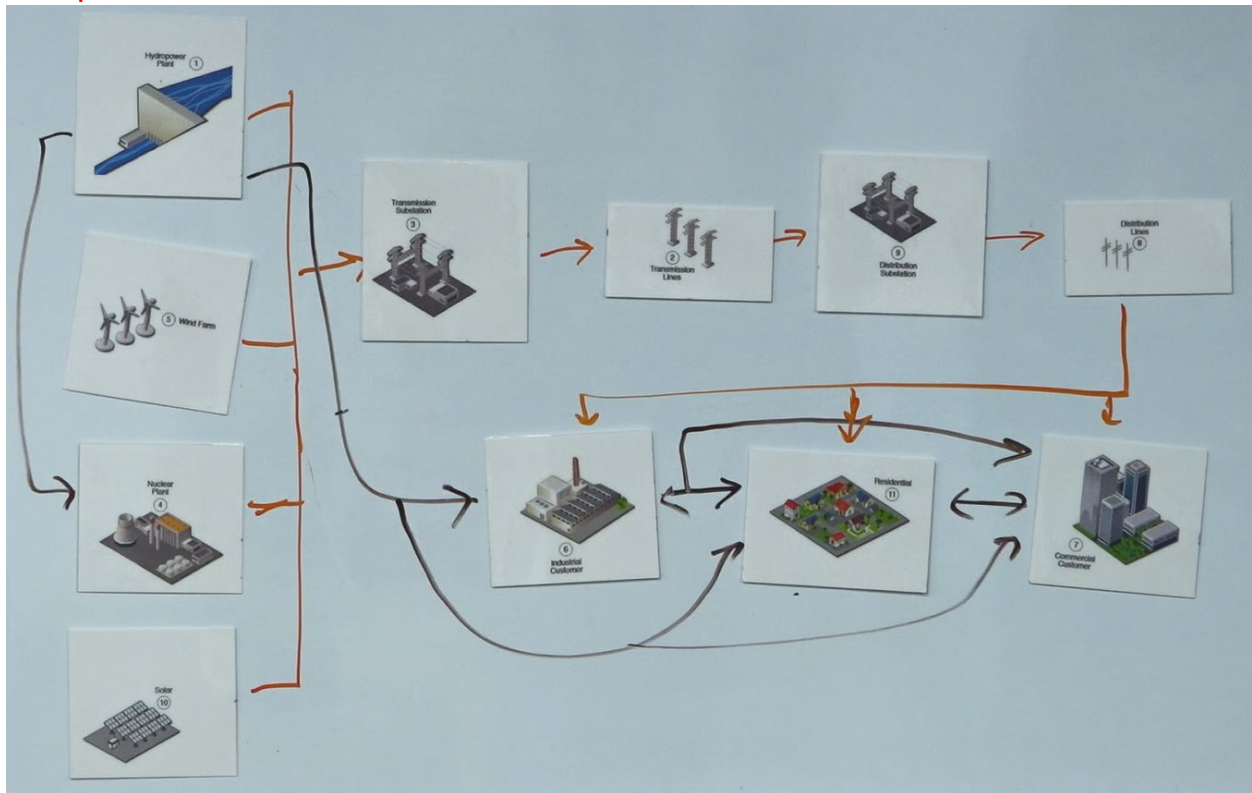
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Student Samples

Below is a list of student samples from a class. Use them to compare against your own drawing to see different approaches to identifying dependencies and power flow.

Sample 1



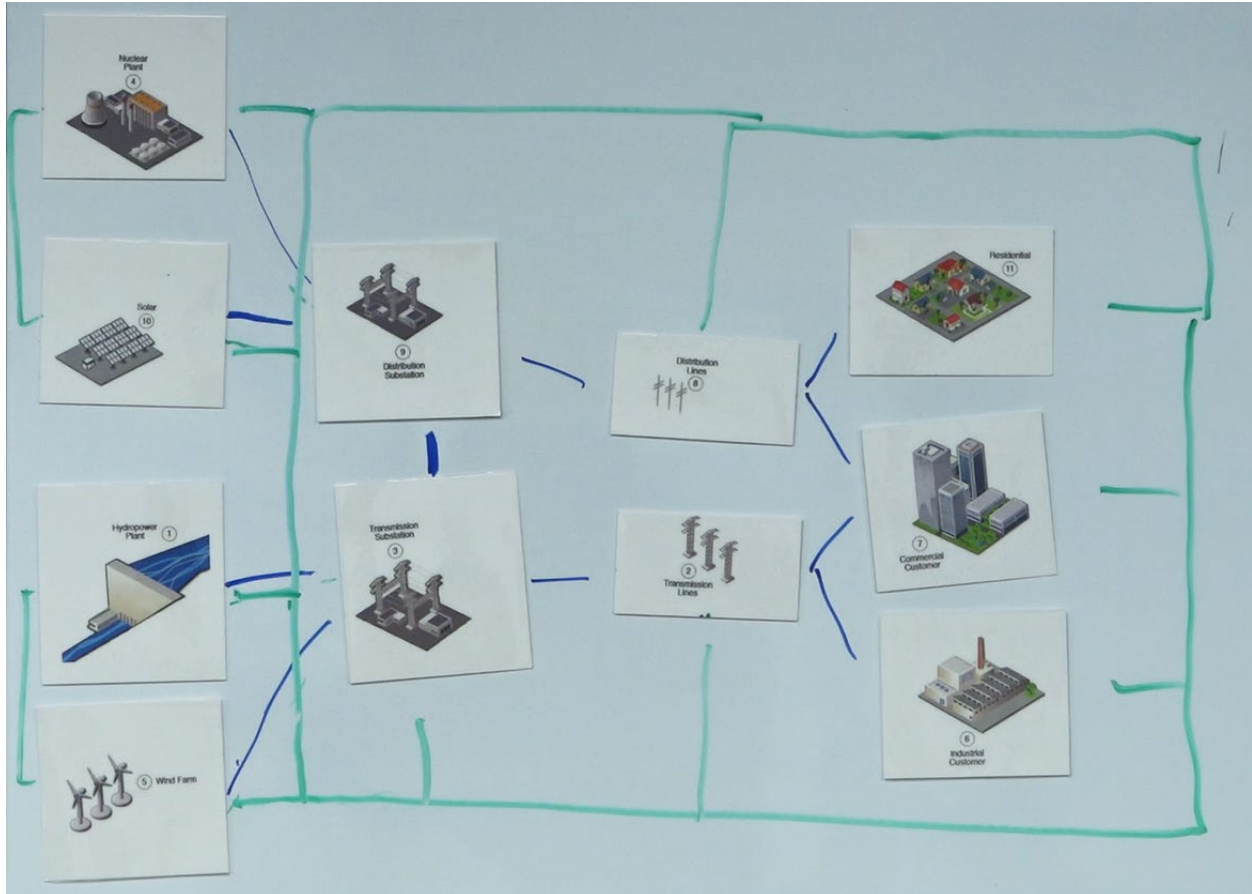
Orange indicates power flow

Brown indicates dependency relationships between components

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Sample 2



Blue indicates power flow

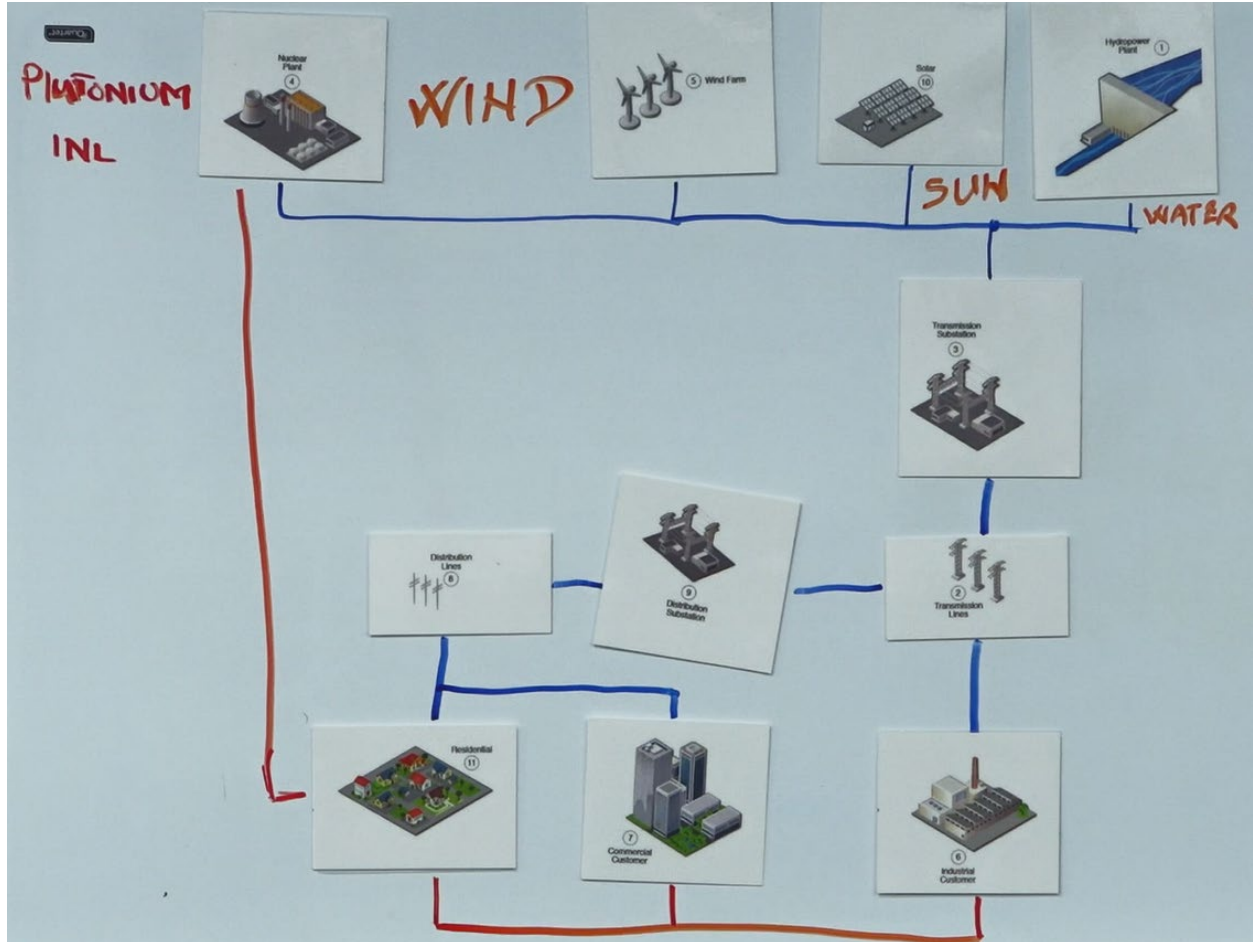
Green indicates dependency relationship between components

Group 2 indicated with their system map that the recipients of power provide raw materials and other resources to the power sources, showing a dependency relationship between them.

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Sample 3



Blue indicates power flow

Orange indicates dependency relationship between components

Group 3 not only highlighted the dependency relationships but identified them by writing them out.